

Gaspard BEAUDOUIN

Machine Learning Research

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Interests: World Models, Multimodal Models, Flow Matching, Representation Learning, Diffusion Language Models.

EXPERIENCE

Research Scientist intern - Mistral AI

Paris, France

Multimodal team

2026

- Conduct research on Multimodal Large Language Models and Vision Encoders. Improving the architecture, dataloading efficiency, and making scaling laws.

Research Intern - Harvard University

Boston, MA, USA

Harvard AI and Robotics Lab, MEE

Feb 2025 – Aug 2025

- Conducted research under Dr. M. Wang on generative modelling and flow matching. Proposed DRFS, an inversion-free flow matching framework for text-to-image editing (*CVPR 2026*), and co-authored *SplitFlow* (*NeurIPS 2025*) for flow aggregation.
- Co-authored *PAGE-4D* (*ICLR 2026*) for pose and geometry estimation of dynamic scenes.

Machine Learning Engineer Intern - Sinequa

Paris, France

Research Team

Aug 2024 – Jan 2025

- Studied LLMs and tool-use agents through supervised fine-tuning and reinforcement learning; designed evaluations for function-calling models using BFCL, data augmentation, and templating.
- Performed distributed instruction fine-tuning with LoRA and full fine-tuning, as well as DPO, on H100 clusters for tool-use models; used Docker and Slurm workflows.

PUBLICATIONS

Delta Velocity Rectified Flow for Text-to-Image Editing *Accepted at CVPR 2026.*

G. Beaudouin, M. Li, J. Kim, S.-H. Yoon, M. Wang

PAGE-4D: Disentangled Pose and Geometry for 4D Perception *Accepted at ICLR 2026 .*

K. Zhou, Y. Wang, G. Chen, G. Beaudouin, F. Zhan, P. Liang, M. Wang

SplitFlow: Flow Decomposition for Inversion-Free Editing *Accepted at NeurIPS 2025.*

S.-H. Yoon, M. Li, G. Beaudouin, C. Wen, M. Azhar, M. Wang

EDUCATION

École Normale Supérieure Paris-Saclay

Paris, France

MVA Master - Mathematics, Vision, Learning

2025 – 2026

- Coursework: Advanced Learning for Text and Graphs, Deep Learning, Optimal Transport, Probabilistic Graphical Models, Large-Scale LLM Training, Generative Modelling, Representation Learning, Stochastic Calculus, Reinforcement Learning, LLMs for Code and Proof.

École Nationale des Ponts et Chaussées - Institut Polytechnique de Paris

Paris, France

Mathematics and Computer Science Department

2022 – 2026

- Focus: Deep Learning, Machine Learning, Computer Vision, Statistics, Convex Optimization, Stochastic Processes, Advanced Programming, Control Theory, and Functional Analysis.

Fénelon Sainte-Marie

Paris, France

Preparatory Classes in Mathematics and Physics

2020 – 2022

- 1st year: PCSI; 2nd year: PSI* with focus on Physics, Mathematics, and Engineering Sciences.

PROJECTS

Native Sparse Attention Reproduction

2026

Academic Project, Large-Scale LLM Training

- Reproduced DeepSeek-AI's Native Sparse Attention with compression, selection, and sliding-window branches; implemented a PyTorch reference transformer with RoPE, RMSNorm, SwiGLU, and GQA-style attention.
- Benchmarked long-context attention on A100-SXM4-40GB, reaching up to $8.3\times$ forward and $3.36\times$ backward speedup at 64k tokens; designed French Needle-in-a-Haystack experiments where NSA reached 97.6% accuracy.

Flow Matching Generalization and Memorization

2026

Academic Project, Generative Modelling

- Reproduced and extended results on closed-form Flow Matching, studying why finite-data optimal velocity fields memorize while neural networks generalize.

- Derived the Flow Matching / diffusion probability-flow ODE equivalence and validated $t_C \sim \log(n)/d$ through dimension, capacity, geometry, and Exact Flow Matching ablations.

LeJEPA Re-implementation and Analysis

2025

Academic Project

- Re-implemented LeJEPA using a Tiny ViT implemented from scratch in PyTorch (17M parameters); studied training dynamics and online linear probing behavior.

Mini GRPO

2025

Personal Project

- Reimplemented GRPO, introduced by DeepSeek-R1 and DeepSeekMath, and applied it to math reasoning tasks.

GPT Transformer from Scratch

2024

Personal Project

- Implemented and trained a PyTorch Mixture-of-Experts Transformer to generate French poetry with recent GPT-OSS techniques; generated synthetic training data and wrote notes on LLMs: From Pre-Training to RLHF (Draft).

Drug Design with Diffusion Models

2024

Academic Project, École des Ponts

- Implemented DDPM and explored DiffDock for drug binding discovery with Sanofi. Poster available.

Edge Detection in Images

2022

Academic Project, École des Ponts

- C++ research project on perceptual boundary saliency; implemented edge detection algorithms.

SKILLS

Languages French native, English fluent, Spanish proficient

Technical Python, PyTorch, Transformers, Docker, Slurm, C++, SQL, R, L^AT_EX

ACADEMIC SERVICE

Conference Reviewing Reviewer for CVPR 2026

MISCELLANEOUS

Sports École des Ponts football team; competitive tennis

Interests Passionate about machine learning research; actively following scientific publications